

# Singularity of Origin: a DNS rebinding attack framework

**Singularity of Origin: a DNS rebinding attack framework** - nccgroup, GitHub.

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## Content

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Readme.md

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## Singularity of Origin

Recent updates:

- **NEW (2025-11-25):** Chrome release 142 added Local Network Access (LNA), which prevents DNS rebinding attack techniques implemented in Singularity. Singularity of Origin's [LNA-from-Non-Secure-Contexts](#) branch implements support for [Trial for Local Network Access from Non-Secure Contexts](#). This temporarily allows access to resources on local networks from non-secure contexts, thus letting you experiment with DNS rebinding attacks a bit longer (until May 18, 2026) when using Chrome.
- **NEW (2025-05-19):** Our demo site, [rebind.it](#), has been upgraded to support IPv6 ([commit 3c5ee9a](#)). This is a breaking change for manual queries (outside of Singularity Manager UI) because the DNS query format has been modified. Please refer to [How to Create Manual DNS Requests to Singularity of Origin](#) for detailed instructions. Additionally, we periodically add new attack payloads; be sure to check them out.

- (2023-04-27) Check out our blog post documenting the [state of DNS rebinding for April 2023](#). We describe Local Network Access, a new draft W3C specification currently implemented in some browsers that aims to prevent DNS rebinding, and show two ways to bypass these restrictions. We also discuss the effects of WebRTC IP address leak mitigation, and DNS Bit 0x20 on DNS rebinding attacks.
- (2020-03-30) New blog post investigating the [impact of DoH on DNS rebinding attacks](#). TL;DR: DoH (DNS over HTTPS) has no effect on rebinding attacks and protections advertised by providers can be bypassed.
- Check out our [DEF CON 27 video](#) and BSidesLV presentation at [State of DNS Rebinding: Attack & Prevention Techniques and

the Singularity of Origin]

(<https://media.defcon.org/DEF%20CON%2027/DEF%20CON%2027%20presentations/DEFCON-27-Gerald-Doussot-Roger-Meyer-State-of-DNS-Rebinding-Attack-and-Prevention-Techniques-and-the-Singularity-of-Origin.pdf>).

`Singularity of Origin` is a tool to perform [DNS rebinding](#) attacks. It includes the necessary components to rebind the IP address of the attack server DNS name to the target machine's IP address and to serve attack payloads to exploit vulnerable software on the target machine.

It also ships with sample payloads to exploit several vulnerable software versions, from the simple capture of a home page to performing remote code execution. It aims at providing a framework to facilitate the exploitation of software vulnerable to DNS rebinding attacks and to raise awareness on how they work and how to protect from them.

Detailed documentation is on the [wiki pages](#).

## Core Features

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- Singularity provides a complete DNS rebinding attack delivery stack:
  - Custom **DNS server** to rebind DNS name and IP address
  - **HTTP server** (manager web interface) to serve HTML pages and JavaScript code to targets and to manage the attacks
  - Several **sample attack payloads**, ranging from grabbing the home page of a target application to performing remote code execution. These payloads can be easily adapted to perform new and custom attacks.
  - Supports DNS CNAME values in target specification in addition to IP addresses to evade DNS filtering solutions or to target internal resources for which the IP address is unknown.
- A simple, fast and efficient HTTP **port scanner** to identify vulnerable services.
- **Attack automation** allows to completely automate the scanning and exploitation of vulnerable services on a network.
- **Hook and Control** permits using victim web browsers as HTTP proxies to access internal network resources, to interactively explore and exploit otherwise inaccessible

applications with your own browser.

## Singularity Manager Interface

The screenshot displays the Singularity Manager Interface in a web browser. The page title is "Singularity of Origin DNS Rebinding Attack". The interface includes fields for "Attack Host" (d.rebind.it), "Attack Host Domain" (35.185.206.165), "Target Port" (3000), and "Attack Payload" (automatic). A "Start Attack" button is visible. A calculator overlay is present in the center, showing a multiplication of 8 by 16, resulting in 128. The calculator also displays a binary representation of the result. The background shows a console window with various log messages, including "Trying payload: pyethapp", "Payload: pyethapp has", "Payload: Rails Console RCE", and "Fetch finished loading".

## Hook and Control a Vulnerable Application on Localhost or Other Hosts

The diagram illustrates the workflow for hooking and controlling a vulnerable application. It starts with the "Singularity of Origin DNS Rebinding Attack" interface, which is configured with "Attack Host" (35.185.206.165), "Target Host" (127.0.0.1), and "Target Port" (8080). The "Attack Payload" is set to "Hook and Control". An arrow points from this interface to a "Hooked WS Clients" window, which lists a client with IP 1996957243. Another arrow points from the "Hooked WS Clients" window to a Jenkins dashboard, which displays "Welcome to Jenkins!" and "Please create new jobs to get started." The Jenkins dashboard also shows a "Build Queue" and "Build Executor Status".

## Automate the Scan and Compromise of All Vulnerables Applications

Singularity of Origin DNS Rebind
+

Not Secure
rebind.it:8080/autoattack.html

The home page of vulnerable services will be dumped in the browser developer console.

### Scanning Progress

```

{ "error": false, "start": 1555711300405, "end": 1555711300453, "duration": 48, "target": { "address": "127.0.0.1", "port": 8080 } }
{ "error": false, "start": 1555711300580, "end": 1555711300616, "duration": 36, "target": { "address": "0.0.0.0", "port": 8080 } }
{ "error": false, "start": 1555711300612, "end": 1555711300621, "duration": 9, "target": { "address": "[REDACTED]", "port": "80" } }
{ "error": false, "start": 1555711300700, "end": 1555711300711, "duration": 11, "target": { "address": "192.168.11.1", "port": "80" } }
{ "error": false, "start": 1555711302425, "end": 1555711302453, "duration": 28, "target": { "address": "192.168.11.11", "port": 8080 } }
{ "error": false, "start": 1555711304995, "end": 1555711305077, "duration": 82, "target": { "address": "192.168.11.21", "port": 8080 } }

```

### DNS Rebinding Progress

|                                                                                                                          |                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>Rebinding...</b><br>target: 127.0.0.1:8080, session: 2558974437, strategy: fs. This page is waiting for a DNS update. | <b>Simple Fetch Get</b><br>target: 0.0.0.0:8080, session: 2317387190, strategy: ma. DNS rebinding successful!             |
| <b>Rebinding...</b><br>target: [REDACTED], session: 2238189191, strategy: fs. This page is waiting for a DNS update.     | <b>Rebinding...</b><br>target: 192.168.11.1:80, session: 3034879025, strategy: fs. This page is waiting for a DNS update. |
| <b>Rebinding...</b>                                                                                                      | <b>Rebinding...</b>                                                                                                       |

Elements
Console
Sources
Network

top
Filter
All levels

☐ Hide network
☒ Log XMLHttpRequests
☒ Preserve log
☐ Eager evaluation
☐ Selected context only
☒ Autocomplete from history
☒ Group similar

```

completion"></div><script>createSearchBox("/search/");</script>
</div></form></div></div><div id="breadcrumbBar"><tr id="top-na
Show more (24.6 KB) Copy

```

15:01:45.136 attack frame s-35.185.206.165-0.0.0.0-2317387190-ma-e.d.rebind.it received message stop payload.js:20
15:02:01.110 DNS rebinding did not happen yet: s-35.185.206.165-127.0.0.1-2558974437-fs-e.d.rebind.it:8080 payload.js:118
15:02:01.215 DNS rebinding did not happen yet: s-35.185.206.165-[REDACTED]-2238189191-fs-e.d.rebind.it payload.js:118
15:02:01.220 DNS rebinding did not happen yet: s-35.185.206.165-192.168.11.1-3034879025-fs-e.d.rebind.it payload.js:118
15:02:02.965 DNS rebinding did not happen yet: s-35.185.206.165-192.168.11.11-2376210315-fs-e.d.rebind.it:8080 payload.js:118
15:02:05.351 Message received from: http://s-35.185.206.165-192.168.11.21-1618037412-fs-e.d.rebind.it:8080 manager.js:462
15:02:05.351 Iframe reports that attack has started: http://s-35.185.206.165-192.168.11.21-1618037412-fs-e.d.rebind.it:8080 manager.js:471
15:02:05.353 attack frame s-35.185.206.165-192.168.11.21-1618037412-fs-e.d.rebind.it received message payload payload.js:20
15:02:05.354 attack frame s-35.185.206.165-192.168.11.21-1618037412-fs-e.d.rebind.it received message interval payload.js:20
15:02:05.354 attack frame s-35.185.206.165-192.168.11.21-1618037412-fs-e.d.rebind.it received message wsproxyport payload.js:20
15:02:05.355 attack frame s-35.185.206.165-192.168.11.21-1618037412-fs-e.d.rebind.it received message indextoken payload.js:20
15:02:05.355 attack frame s-35.185.206.165-192.168.11.21-1618037412-fs-e.d.rebind.it received message flushdns payload.js:20
15:02:05.355 attack frame s-35.185.206.165-192.168.11.21-1618037412-fs-e.d.rebind.it received message start payload.js:20
15:02:05.356 frame s-35.185.206.165-192.168.11.21-1618037412-fs-e.d.rebind.it waiting 20000 milliseconds for dns update payload.js:49

Console
What's New

## Usage

Setting up Singularity requires a DNS domain name where you can edit your own DNS records for your domain and a Linux server to run it. Please see the [setup singularity](#) wiki page for detailed instructions.

The documentation is on the [wiki pages](#). Here are a few pointers to start:

- What are [DNS Rebinding Attacks](#)?
- [Preventing DNS Rebinding Attacks](#)
- [Setup and Installation](#)
- Description of existing [Payloads](#) and how to write your own

A test instance is available for demo purposes at <http://rebind.it:8080/manager.html>.

## Speed

Singularity has been tested to work with the following browsers in optimal conditions in under **3 seconds**:

| Browser    | Operating System | Time to Exploit | Rebinding Strategy            | Fetch Interval | Target Specification |
|------------|------------------|-----------------|-------------------------------|----------------|----------------------|
| ~~Chrome~~ | ~~Windows 10~~   | <s>~3s</s>      | ~~ Multiple answers (fast) ~~ | ~~1s~~         | ~~127.0.0.1~~        |

| Browser  | Operating System | Time to Exploit | Rebinding Strategy            | Fetch Interval | Target Specification |
|----------|------------------|-----------------|-------------------------------|----------------|----------------------|
| ~~Edge~~ | ~~Windows 10~~   | <s>~3s</s>      | ~~ Multiple answers (fast) ~~ | ~~1s~~         | ~~127.0.0.1~~        |
| Firefox  | Windows 10       | ~3s             | Multiple answers (fast)       | 1s             | 127.0.0.1            |
| Chromium | Ubuntu           | ~3s             | Multiple answers (fast)       | 1s             | 0.0.0.0              |
| Firefox  | Ubuntu           | ~3s             | Multiple answers (fast)       | 1s             | 0.0.0.0              |
| Chrome   | macOS            | ~3s             | Multiple answers (fast)       | 1s             | 0.0.0.0              |
| Firefox  | macOS            | ~3s             | Multiple answers (fast)       | 1s             | 0.0.0.0              |
| Safari   | macOS            | ~3s             | Multiple answers (fast)       | 1s             | 0.0.0.0              |

## Payloads Description

Singularity supports the following attack payloads:

- **Basic fetch request** ( `simple-fetch-get.js` ): This sample payload

makes a GET request to the root directory ('/') and shows the server response using the `fetch` API. The goal of this payload is to function as example request to make additional contributions as easy as possible.

- **automatic**: This payload automatically attempts to detect known services and exploit them using other payloads listed in this section or that were developed and added to Singularity by users.

- **Chrome DevTools RCE** ( `exposed-chrome-devtools.js` ): This payload

demonstrates a remote code execution (RCE) vulnerability in Microsoft VS Code fixed in version 1.19.3. This payload can be adapted to exploit any software that exposes Chrome Dev Tools on `localhost`.

- **Etcd k/v dump** ( `etcd.js` ): This payload retrieves the keys and values from the `etcd` key-value store.

- **pyethapp** ( `pyethapp.js` ): Exploits the Python implementation of the Ethereum client `Pyethapp` to get the list of owned eth addresses and retrieve the balance of the first eth address.

- **Rails Console RCE** ( `rails-console-rce.js` ): Performs a remote code execution (RCE) attack on the [Rails Web Console](#).

- **AWS Metadata Exfil** ( `aws-metadata-exfil.js` ): Forces a headless browser to exfiltrate AWS metadata

including private keys to a given host. Check the payload contents for additional details on how to setup the attack.

- **Duplicati RCE** ( `duplicati-rce.js` ): This payload exploits the

Duplicati backup client and performs a remote code execution (RCE) attack. For this attack to work, parameter `targetURL` in file `payload-duplicati-rce.html` must be updated to point to a valid Duplicati backup containing the actual RCE payload, a shell script.

- **WebPDB** ( `webpdb.js` ): A generic RCE payload to exploit `PDB`, a python debugger exposed via websockets.

- **Hook and Control** ( `hook-and-control.js` ): Hijack target browsers and use them to access inaccessible resources from your own browser or other HTTP clients. You can retrieve the list of hooked browsers on the "soohooked" sub-domain of the Singularity manager host on port 3129 by default e.g. `http://soohooked.rebinder.your.domain:3129/`. To authenticate, submit the secret value dumped to the console by the Singularity server at startup.

- **Jenkins Script Console** ( `jenkins-script-console.js` ): This payload exploits the [Jenkins Script Console](#) and displays the stored credentials.

- **Docker API** ( `docker-api.js` ): This payload exploits the [Docker API](#) and displays the `/etc/shadow` file of the Docker host.

- **Ollama Llama2 Exfil** ( `ollama-exfil.js` ): Exfiltrate files from hosts running Ollama, an open-source system for running and managing large language models (LLMs). See [blog post](#).
- And more, check the payloads folder ( `html/payloads` ).